

A Finite Periodic Extension of an Aperiodic Minimal Subshift

Route-A structural certificate C149

Abstract

We adjoin tagged cycles of lengths 1, 2, 3, 5 to the aperiodic minimal Thue–Morse subshift. The resulting system has four primitive cycles and rational Artin–Mazur zeta $((1-z)(1-z^2)(1-z^3)(1-z^5))^{-1}$. This supplies a controlled finite periodic skeleton but destroys minimality.

1 The disjoint system

Let X_{TM} be the two-sided subshift generated by $0 \mapsto 01$, $1 \mapsto 10$. It is nonempty and minimal and contains no periodic point. For $E = \{1, 2, 3, 5\}$, let C_ℓ be a tagged cycle of length ℓ and set

$$Y = X_{\text{TM}} \sqcup \bigsqcup_{\ell \in E} C_\ell.$$

The map is the shift on X_{TM} and cyclic successor on each C_ℓ . One iterate is the clock.

2 All-period counts

The component C_ℓ contributes all ℓ of its points to $\text{Fix}(T^n)$ exactly when $\ell \mid n$. Therefore

$$\# \text{Fix}(T^n) = \sum_{\substack{\ell \in E \\ \ell \mid n}} \ell. \tag{1}$$

Each component is a single cycle of its displayed least period. With

$$\zeta_Y(z) = \exp \left(\sum_{n \geq 1} \# \text{Fix}(T^n) \frac{z^n}{n} \right),$$

we obtain

$$\begin{aligned} \log \zeta_Y(z) &= \sum_{\ell \in E} \sum_{q \geq 1} \frac{z^{\ell q}}{q} = - \sum_{\ell \in E} \log(1 - z^\ell), \\ \zeta_Y(z) &= \frac{1}{(1-z)(1-z^2)(1-z^3)(1-z^5)}. \end{aligned} \tag{2}$$

3 Minimality obstruction and validation

Every tagged cycle is a proper nonempty closed invariant subset of Y , so Y is not minimal. A 60-period exact ledger and degree-30 zeta series agree with (1)–(2). The independent checker passes 395 assertions, SymPy passes 277 checks, replay is byte-identical, and 41 hostile cases are rejected.

The strict tuple is (A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL). The cycles are declared components, not points of X_{TM} . We claim no target divisor, arithmetic/local factor, root number, automorphy, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.